In [162…
```
from sympy import real_root, Order, latex
from sympy.physics.quantum import Commutator, Operator
```

In [2]:
```
#Define the symbols s and e
from sympy.abc import s,e

#Define Operators T and V
T = Operator('T')
V = Operator('V')
```

Baker-Campbell-Hausdorf formula

$$e^{\epsilon(\hat{T}+\hat{V})} = \exp\left(\hat{T}+\hat{V}+\tfrac{1}{2}[\hat{T},\hat{V}]+\cdots\right)$$

In [3]:
```
# Baker-Campbell-Hausdorf formula up to 4 nested commutators
def bch(X, Y, order=1):
    XY = Commutator(X, Y)
    YX = -XY
    out = X + Y + XY/2
    if(order == 1):
        return out

    XXY = Commutator(X,XY)
    YYX = Commutator(Y,YX)
    out = out + (XXY + YYX)/12
    if(order == 2):
        return out

    YXXY = Commutator(Y, XXY)
    out = out - YXXY/24
    if(order == 3):
        return out


    YYYX = Commutator(Y, YYX)
    YYYYX = Commutator(Y, YYYX)
    XYYYX = Commutator(X, YYYX)
    XXXY = Commutator(X, XXY)
    XXXXY = Commutator(X, XXXY)
    YXXXY = Commutator(Y, XXXY)
    YXY = Commutator(Y, XY)
    XYXY = Commutator(X, YXY)
    YXYXY = Commutator(Y, XYXY)
    XYX = Commutator(X, YX)
    YXYX = Commutator(Y, XYX)
    XYXYX = Commutator(X, YXYX)

    out = out - (YYYYX + XXXXY)/720 + (XYYYX + YXXXY)/360 + (YXYXY + XYXYX)/12(
    if(order == 4):
        return out
    print("Not Implemented: BCH not evaluated on order %d\nReverting to order 4
    return out
```

In [4]:
```
bch(T,V, 2)
```





Out[4]:

$$\frac{-[[T,V],T]+[[T,V],V]}{12}+\frac{[T,V]}{2}+T+V$$

## Operator $\hat{T}^{(1)}_{\mathrm{a}}$

$\hat{T}_{1\mathrm{a}} = e^{\epsilon\hat{V}}e^{\epsilon\hat{T}}$

In [178…
```
def t1_a(e, order=1):
    out = bch(e*V, e*T, order=order)
    for i in range(order):
        out = out.expand(commutator=True)
    return out
```

In [179…
```
t1_a(e, order=4) + Order(e**4)
```

Out[179]:

$$eV+eT-\frac{e^2[T,V]}{2}+\frac{e^3[[T,V],V]}{12}-\frac{e^3[[T,V],T]}{12}+O\left(e^4\right)$$

In [180…
```
t1_a(e, order=2) + Order(e**4)
```

Out[180]:

$$eV+eT-\frac{e^2[T,V]}{2}+\frac{e^3[[T,V],V]}{12}-\frac{e^3[[T,V],T]}{12}+O\left(e^4\right)$$

## Operator $\hat{T}^{(1)}_{\mathrm{b}}$

$\hat{T}_{1\mathrm{b}} = e^{\epsilon\hat{T}}e^{\epsilon\hat{V}}$

In [181…
```
def t1_b(e, order=1):
    out = bch(e*T, e*V, order=order)
    for i in range(order):
        out = out.expand(commutator=True)
    return out
```

In [182…
```
t1_b(e, order=4) + Order(e**4)
```

Out[182]:

$$eV+eT+\frac{e^2[T,V]}{2}+\frac{e^3[[T,V],V]}{12}-\frac{e^3[[T,V],T]}{12}+O\left(e^4\right)$$

In [169…
```
t1_b(e, order=2) + Order(e**4)
```

Out[169]:

$$eV+eT+\frac{e^2[T,V]}{2}+\frac{e^3[[T,V],V]}{12}-\frac{e^3[[T,V],T]}{12}+O\left(e^4\right)$$

## Operator $\hat{T}^{(2)}_{\mathrm{PV}}$

$\hat{T}_{\rm PV} = \hat{T}_{\rm 1b}\left(\frac{\epsilon}{2}\right)\hat{T}_{\rm 1a}\left(\frac{\epsilon}{2}\right)$

In [126…
```
def t2_pv(e, order_t1=1, order_bch=1):
    o1 = t1_b(e/2, order=order_t1)
    o2 = t1_a(e/2, order=order_t1)
    out = bch(o1, o2, order=order_bch)
    for i in range(order_t1+1):
        out = out.expand(commutator=True)
    return out
```

In [153…
```
t2_pv(e, order_t1=3, order_bch=1) + Order(e**5)
```

Out[153]:

$$eV + eT + \frac{e^3\left[\left[T,V\right],V\right]}{12} + \frac{e^3\left[\left[T,V\right],T\right]}{24} + \frac{e^4\left[\left[\left[T,V\right],V\right],T\right]}{384} - \frac{e^4\left[\left[\left[T,V\right],T\right],V\right]}{384} +$$

In [154…
```
t2_pv(e, order_t1=3, order_bch=3) + Order(e**5)
```

Out[154]:

$$eV + eT + \frac{e^3\left[\left[T,V\right],V\right]}{12} + \frac{e^3\left[\left[T,V\right],T\right]}{24} + \frac{e^4\left[\left[\left[T,V\right],V\right],T\right]}{384} - \frac{e^4\left[\left[\left[T,V\right],T\right],V\right]}{384} +$$

In [166…
```
T2_PV = t2_pv(e, order_t1=3, order_bch=3) + Order(e**5)
print(latex(T2_PV))
```

```
e V + e T + \frac{e^{3} \left[\left[T,V\right],V\right]}{12} + \frac{e^{3}
\left[\left[T,V\right],T\right]}{24} + \frac{e^{4} \left[\left[\left[T,V\rig
ht],V\right],T\right]}{384} - \frac{e^{4} \left[\left[\left[T,V\right],T\rig
ht],V\right]}{384} + O\left(e^{5}\right)
```

## Operator $\hat{T}^{(2)}_{\rm VV}$

$\hat{T}_{\rm VV} = \hat{T}_{\rm 1a}\left(\frac{\epsilon}{2}\right)\hat{T}_{\rm 1b}\left(\frac{\epsilon}{2}\right)$

In [135…
```
def t2_vv(e, order_t1=1, order_bch=1):
    o1 = t1_a(e/2, order=order_t1)
    o2 = t1_b(e/2, order=order_t1)
    out = bch(o1, o2, order=order_bch)
    for i in range(order_t1+1):
        out = out.expand(commutator=True)
    return out
```

In [152…
```
t2_vv(e, order_t1=3, order_bch=1) + Order(e**5)
```

Out[152]:

$$eV + eT - \frac{e^3\left[\left[T,V\right],V\right]}{24} - \frac{e^3\left[\left[T,V\right],T\right]}{12} + \frac{e^4\left[\left[\left[T,V\right],V\right],T\right]}{384} - \frac{e^4\left[\left[\left[T,V\right],T\right],V\right]}{384} +$$

In [137…
```
t2_vv(e, order_t1=3, order_bch=3) + Order(e**5)
```





Out[137]:

$$eV + eT - \frac{e^3\left[\left[T,V\right],V\right]}{24} - \frac{e^3\left[\left[T,V\right],T\right]}{12} + \frac{e^4\left[\left[\left[T,V\right],V\right],T\right]}{384} - \frac{e^4\left[\left[\left[T,V\right],T\right],V\right]}{384} +$$

## Operator $\hat{T}^{(4)}_{\rm PV}$

$\hat{T}_4 = \hat{T}_{\rm PV}(\epsilon)\hat{T}_{\rm PV}(-s\epsilon)\hat{T}_{\rm PV}(\epsilon)$

$s = 2^{1/3}$

In [138…
```
cbrt_2 = real_root(2,3)
cbrt_2
```

Out[138]: $\sqrt[3]{2}$

In [155…
```
def t4_pv(e, order_bch=1, order_t2=1, order_t1=3):
    o1 = t2_pv(e,         order_bch=order_t2, order_t1=order_t1)
    o2 = t2_pv(-cbrt_2*e, order_bch=order_t2, order_t1=order_t1)
    o3 = t2_pv(e,         order_bch=order_t2, order_t1=order_t1)

    o21 = bch(o2, o1, order=order_bch)
    for i in range(order_t2+1):
        o21 = o21.expand(commutator=True)
    o321 = bch(o3, o21,   order=order_bch)
    for i in range(order_t2+1):
        o321 = o321.expand(commutator=True)

    return o321
```

In [160…
```
t4_pv(e, order_t2=1, order_t1=1) + Order(e**7)
```

Out[160]:

$$2eV - \sqrt[3]{2}eV + 2eT - \sqrt[3]{2}eT + \frac{e^5\left[\left[\left[\left[T,V\right],V\right],V\right],V\right]}{32} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],V\right],V\right],V\right]}{64} +$$
$$- \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],V\right],T\right],V\right]}{64} + \frac{e^5\left[\left[\left[\left[T,V\right],V\right],T\right],T\right]}{32} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],V\right],T\right],T\right]}{64} +$$
$$- \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],T\right],V\right],T\right]}{64} + \frac{e^5\left[\left[\left[\left[T,V\right],T\right],T\right],V\right]}{32} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],T\right],T\right],V\right]}{64} +$$

In [161…
```
t4_pv(e, order_t2=1, order_t1=2) + Order(e**7)
```

Out[161]:

$$2eV - \sqrt[3]{2}eV + 2eT - \sqrt[3]{2}eT + \frac{2^{\frac{2}{3}}e^5\left[\left[\left[T,V\right],V\right],\left[T,V\right]\right]}{384} - \frac{e^5\left[\left[\left[T,V\right],V\right],\left[T,V\right]\right]}{384} + \cdots$$
$$- \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],V\right],V\right],V\right]}{48} + \frac{e^5\left[\left[\left[\left[T,V\right],V\right],V\right],T\right]}{24} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],V\right],V\right],T\right]}{48} + \cdots$$
$$- \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],V\right],T\right],T\right]}{48} + \frac{e^5\left[\left[\left[\left[T,V\right],T\right],V\right],V\right]}{48} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],T\right],V\right],V\right]}{96} + \cdots$$
$$- \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],T\right],T\right],V\right]}{96} + \frac{e^5\left[\left[\left[\left[T,V\right],T\right],T\right],T\right]}{48} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],T\right],T\right],T\right]}{96} + \cdots$$

In [157…

```
t4_pv(e) + Order(e**7)
```

Out[157]:

$$2eV - \sqrt[3]{2}eV + 2eT - \sqrt[3]{2}eT + \frac{\sqrt[3]{2}e^4\left[\left[\left[T,V\right],V\right],T\right]}{192} + \frac{e^4\left[\left[\left[T,V\right],V\right],T\right]}{192} - \cdots$$
$$+ \frac{e^5\left[\left[\left[T,V\right],T\right],\left[T,V\right]\right]}{384} - \frac{2^{\frac{2}{3}}e^5\left[\left[\left[T,V\right],T\right],\left[T,V\right]\right]}{384} + \frac{e^5\left[\left[\left[\left[T,V\right],V\right],V\right],V\right]}{24} - \cdots$$
$$+ \frac{31e^5\left[\left[\left[\left[T,V\right],V\right],T\right],V\right]}{768} + \frac{2^{\frac{2}{3}}e^5\left[\left[\left[\left[T,V\right],V\right],T\right],V\right]}{768} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],V\right],T\right],V\right]}{48} + \cdots$$
$$+ \frac{5e^5\left[\left[\left[\left[T,V\right],T\right],V\right],V\right]}{256} + \frac{2^{\frac{2}{3}}e^5\left[\left[\left[\left[T,V\right],T\right],V\right],V\right]}{768} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],T\right],V\right],V\right]}{96} + \cdots$$
$$+ \frac{e^5\left[\left[\left[\left[T,V\right],T\right],T\right],V\right]}{48} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],T\right],T\right],V\right]}{96} + \frac{e^5\left[\left[\left[\left[T,V\right],T\right],T\right],T\right]}{48} - \cdots$$
$$- \frac{\sqrt[3]{2}e^6\left[\left[\left[\left[\left[T,V\right],V\right],T\right],V\right],V\right]}{512} - \frac{\sqrt[3]{2}e^6\left[\left[\left[\left[\left[T,V\right],V\right],T\right],V\right],T\right]}{512} - \cdots$$
$$+ \frac{\sqrt[3]{2}e^6\left[\left[\left[\left[\left[T,V\right],T\right],V\right],V\right],T\right]}{512} + \frac{\sqrt[3]{2}e^6\left[\left[\left[\left[\left[T,V\right],T\right],V\right],T\right],V\right]}{512} + \cdots$$

In [183…

```
t4_pv(e, order_t2=2) + Order(e**7)
```





Out[183]:

$$2eV - \sqrt[3]{2}eV + 2eT - \sqrt[3]{2}eT + \frac{\sqrt[3]{2}e^4\left[\left[\left[T,V\right],V\right],T\right]}{192} + \frac{e^4\left[\left[\left[T,V\right],V\right],T\right]}{192} - \cdots$$
$$+ \frac{2^{\frac{2}{3}}e^5\left[\left[\left[T,V\right],T\right],\left[T,V\right]\right]}{384} - \frac{e^5\left[\left[\left[T,V\right],T\right],\left[T,V\right]\right]}{384} + \frac{e^5\left[\left[\left[\left[T,V\right],V\right],V\right],V\right]}{24} - \cdots$$
$$+ \frac{31e^5\left[\left[\left[\left[T,V\right],V\right],T\right],V\right]}{768} + \frac{2^{\frac{2}{3}}e^5\left[\left[\left[\left[T,V\right],V\right],T\right],V\right]}{768} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],V\right],T\right],V\right]}{48} + \cdots$$
$$+ \frac{5e^5\left[\left[\left[\left[T,V\right],T\right],V\right],V\right]}{256} + \frac{2^{\frac{2}{3}}e^5\left[\left[\left[\left[T,V\right],T\right],V\right],V\right]}{768} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],T\right],V\right],V\right]}{96} + \cdots$$
$$+ \frac{e^5\left[\left[\left[\left[T,V\right],T\right],T\right],V\right]}{48} - \frac{\sqrt[3]{2}e^5\left[\left[\left[\left[T,V\right],T\right],T\right],V\right]}{96} + \frac{e^5\left[\left[\left[\left[T,V\right],T\right],T\right],T\right]}{48} - \cdots$$
$$- \frac{\sqrt[3]{2}e^6\left[\left[\left[\left[\left[T,V\right],V\right],T\right],V\right],V\right]}{512} - \frac{\sqrt[3]{2}e^6\left[\left[\left[\left[\left[T,V\right],V\right],T\right],V\right],T\right]}{512} - \cdots$$
$$+ \frac{\sqrt[3]{2}e^6\left[\left[\left[\left[\left[T,V\right],T\right],V\right],V\right],T\right]}{512} + \frac{\sqrt[3]{2}e^6\left[\left[\left[\left[\left[T,V\right],T\right],V\right],T\right],V\right]}{512} + \cdots$$